

Trainings ▾

Preparatory Steps

Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50° C [120° F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

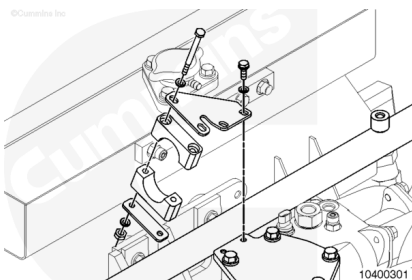
Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Disconnect the coolant vent line from the aftercooler transfer tube. Refer to Procedure 008-017 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>)

Remove

Marine Applications

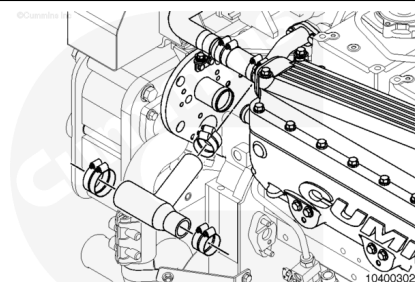
Remove the vibration isolator from the water transfer tube.



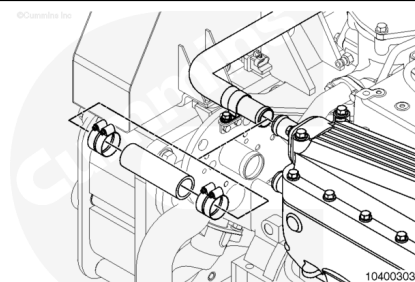
Loosen the six hose clamps from the coolant inlet transfer connection.

Remove the transfer connection.

Remove the thermostat housing coolant inlet tube and the orifice plate connected to the thermostat housing, if equipped.

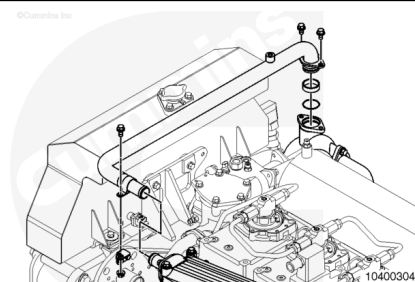


Loosen the four hose clamps and remove the hose from the aftercooler discharge to the transfer tube.



Remove the two capscrews from the transfer tube at the coolant bypass connection and the p-clamp.

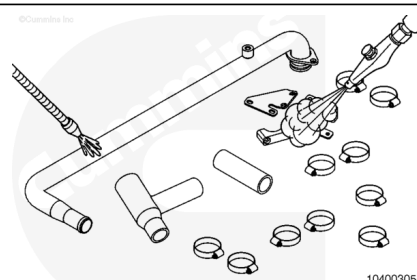
Remove the transfer tube and wear sleeve. Discard the o-ring.



Clean and Inspect for Reuse

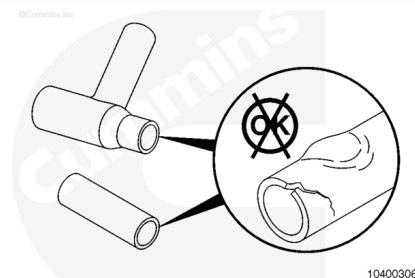
⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use solvent to clean the removed components.

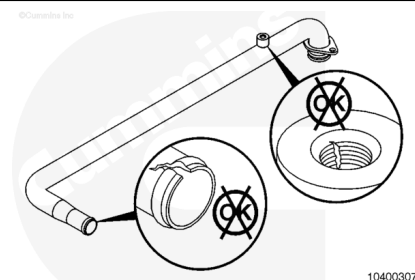
Inspect the hoses for signs of stress and cracking. If damage is found, replace the hoses.



Inspect the transfer tube for damage or signs of leakage.

Check the threads of the vent connection for stripped or damaged threads.

If damage is found, replace the transfer tube.



Install

Marine Applications

⚠ CAUTION ⚠

Do not use petroleum based oil. This will cause swelling of the o-ring.

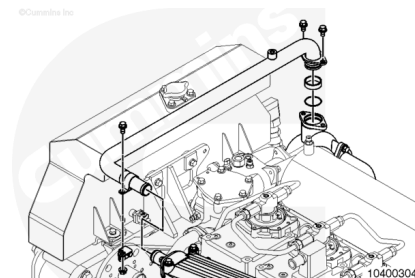
Lubricate the o-ring with vegetable oil and install it onto the transfer tube. Install the wear sleeve, transfer tube and two capscrews into the bypass tube.

Install the p-clamp capscrew.

Torque Value:

Transfer Tube Capscrews

1. 45 n•m [33 ft-lb]

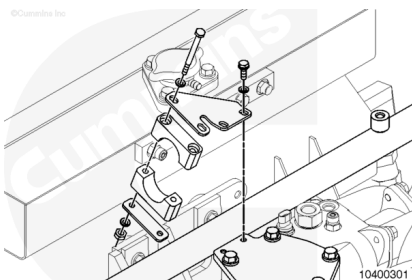


Install the vibration isolator and tighten the capscrews.

Torque Value:

Vibration Isolator Capscrews

1. 45 n•m [33 ft-lb]

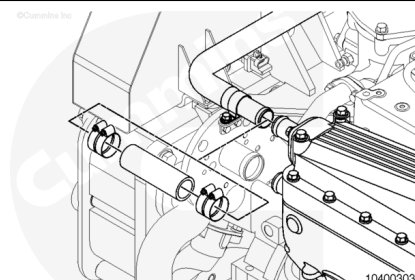


Install the aftercooler discharge hose and clamps. Tighten the clamps.

Torque Value:

Aftercooler Discharge Clamps

1. 20 n•m [177 in-lb]



Install the thermostat housing coolant inlet tube and the orifice plate connected to the thermostat housing, if equipped.

Torque Value:

Transfer Tube Capscrews

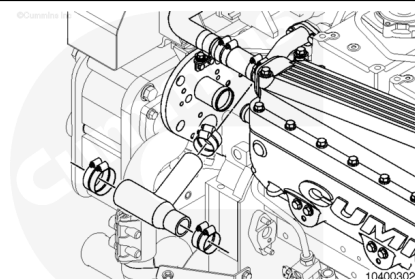
1. 45 n•m [33 ft-lb]

Install the coolant inlet transfer connection and the six hose clamps.

Torque Value:

Coolant Inlet Transfer Connection Clamps

1. 20 n•m [177 in-lb]

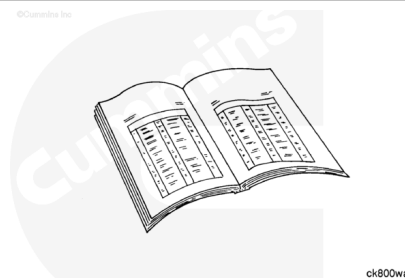


Finishing Steps

Marine Applications

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



⚠ CAUTION ⚠

Use caution when filling the cooling system that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

- Connect the coolant vent line from the aftercooler transfer tube. Refer to Procedure 008-017 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Operate the engine and check for leaks.